

PORT-GENTIL, Gabon

WMO: 645010

Lat: 0.712S

Lon: 8.754E

Elev: 13

StdP: 14.69

Time Zone: 1.00 (E01)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	70.8	71.5	66.2	96.5	74.2	67.4	100.9	74.3	18.5	79.1	17.6	78.7	7.0	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.7	89.6	80.7	88.2	80.5	87.9	80.5	82.1	86.7	81.6	86.2	81.0	85.7	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	159.9	85.9	80.2	157.1	85.4	79.2	151.8	84.5	46.0	87.6	45.2	87.1	44.8	87.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	15.8	14.5	DB	69.1	92.8	0.9	4.1	68.4	95.8	67.9	98.1	67.3	100.4	66.7	103.4	(4)
(5)			WB	65.7	83.9	3.7	0.7	63.1	84.4	60.9	84.8	58.9	85.1	56.2	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.5	82.3	82.7	83.2	82.7	81.6	78.2	75.9	77.2	79.5	80.7	80.7	81.4
	DBStd	2.82	1.77	1.94	1.73	1.69	1.67	2.08	1.85	1.46	1.19	1.34	1.67	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11126	1001	916	1029	980	980	846	802	843	884	951	921	973
	CDD65	5651	536	496	564	530	515	396	337	378	434	486	471	508
	CDH74	54239	6058	5782	6682	6117	5407	2722	1391	2056	3500	4552	4636	5336
CDH80	13930	1895	2033	2495	2139	1418	289	61	102	399	789	970	1340	
Wind	WSAvg	8.6	7.3	7.2	7.1	7.0	8.1	10.1	10.1	10.6	11.0	9.5	8.1	7.1
Precipitation	PrecAvg	75.0	10.0	6.9	10.5	11.4	5.6	0.1	0.0	0.2	1.1	6.9	14.0	9.3
	PrecMax	102.2	17.4	17.1	18.1	21.9	15.4	0.4	1.0	1.1	3.9	17.3	26.3	21.5
	PrecMin	52.5	1.4	2.2	4.6	1.7	0.0	0.0	0.0	0.0	0.0	0.8	6.6	2.3
	PrecStd	13.1	4.0	3.2	3.3	5.6	4.4	0.2	0.2	0.3	0.9	4.4	5.0	5.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	89.8	90.0	90.0	89.3	84.6	82.8	83.9	84.7	86.4	87.5	88.2
		MCWB	81.2	80.8	81.0	80.7	80.5	77.6	75.1	76.1	77.2	78.7	79.8	80.1
	2%	DB	88.0	88.5	89.6	89.3	87.8	84.0	81.5	82.3	84.2	85.7	86.2	87.2
		MCWB	80.6	80.7	80.7	80.5	80.4	77.2	74.3	75.0	76.9	78.3	79.1	79.8
	5%	DB	87.4	87.9	88.2	88.0	86.3	82.6	80.5	80.9	83.0	84.4	85.5	86.1
		MCWB	80.3	80.6	80.5	80.5	79.8	76.2	73.7	74.1	76.2	77.6	78.8	79.3
	10%	DB	86.1	87.2	87.7	87.4	85.6	82.0	79.0	80.3	82.4	83.9	84.3	85.0
		MCWB	79.6	80.3	80.5	80.3	79.5	75.9	73.3	73.9	75.8	77.4	78.2	78.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	82.8	82.7	82.7	81.8	79.2	76.3	77.0	79.0	80.0	80.7	81.2
		MCDB	86.9	87.1	87.3	87.1	85.7	82.9	80.6	81.1	83.0	84.2	84.6	85.5
	2%	WB	81.3	81.9	81.9	81.8	80.9	78.3	75.4	75.9	78.2	79.1	79.9	80.5
		MCDB	85.9	86.6	86.7	86.4	85.3	82.2	79.6	80.0	82.1	83.4	83.9	84.8
	5%	WB	80.7	81.1	81.3	81.1	80.4	77.4	74.6	75.3	77.4	78.6	79.3	79.9
		MCDB	85.3	85.9	86.3	85.8	84.7	81.2	78.8	79.3	81.3	82.8	83.2	84.1
	10%	WB	80.1	80.6	80.8	80.7	79.9	76.6	73.9	74.6	76.7	78.1	78.9	79.4
		MCDB	84.5	85.5	85.9	85.5	84.0	80.5	78.1	78.7	80.5	82.2	82.8	83.4
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	9.7	9.5	7.7	6.8	7.0	6.6	6.8	7.2	7.7	7.8
		MCDBR	8.8	9.2	10.4	10.5	8.9	7.3	7.7	7.1	7.5	7.8	8.6	8.8
		MCWBR	4.2	4.3	5.0	5.0	4.3	3.4	3.4	3.3	3.5	3.6	4.2	4.0
	5% WB	MCDBR	8.4	8.9	9.8	9.8	8.6	6.8	7.4	6.9	7.2	7.5	8.1	8.3
MCWBR		4.5	4.6	5.2	5.3	4.5	3.7	3.4	3.5	3.6	3.8	4.5	4.2	
Clear-Sky Solar Irradiance	taub	0.637	0.619	0.564	0.468	0.512	0.607	0.656	0.631	0.515	0.461	0.497	0.612	
	taud	1.855	1.903	2.060	2.374	2.193	1.902	1.786	1.848	2.178	2.364	2.265	1.906	
	Ebn at Noon	230	237	249	267	248	220	211	223	257	275	265	234	
	Edh at Noon	68	66	56	40	46	60	68	66	49	41	45	64	
All-Sky Solar Radiation	RadAvg	1432	1449	1524	1484	1384	1322	1308	1321	1387	1413	1306	1341	
	RadStd	50	75	67	75	116	90	64	82	71	76	59	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page